

Identifying Long Haplotype Blocks with Low Diversity

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Abstract.

Given an $m \times n$ haplotype matrix A , we show linear time algorithms for finding all interval diversities, farthest sites, and the longest block with low diversity.

For selecting the multiple long blocks with diversity constraint, we show that selecting k blocks with longest total length can be found in $O(nk)$. We also propose linear time algorithms in calculating the all intra-longest-blocks and all intra- k -longest-blocks. For dealing with missing SNP data, we propose methods that clarifies ambiguous SNP sites; furthermore, we show some hardness result concerning the minimum diversity problem.

Keywords: algorithm, SNP, haplotype, diversity, haplotype block selection, missing data.

1 Introduction

Mutation in DNA is the principle factor that is responsible for the phenotypic differences among human beings, and SNPs (single nucleotide polymorphisms) are the most common mutations, hence it is fundamental to complete a map of all SNPs in the human population. For this purpose a SNP is defined as a position in a chromosome where each one of two (or more) specific nucleotides are observed in at least 10% of the population [15]. The nucleotides involved in a SNP are called *alleles*. It has been observed that for almost all SNPs only

two different alleles are present, in such case the SNP is said biallelic, otherwise the SNP is said multiallelic. In this paper, we will consider exclusively biallelic SNPs.

In diploid organisms, such as humans, each chromosome is made of two distinct copies and each copy is called a *haplotype*. It is known that exactly one haplotype is inherited from the father and the other is from the mother. More precisely, in the absence of recombinations events, each haplotype in a child is identical to one of the two haplotypes of each parent. Whenever recombinations occur, a haplotype of the child may consists of portions of both haplotypes of a parent. A current high-priority phase of human genomics involves the development of a full *Haplotype Map* of the human genome [11].

There is currently a great deal of interest in how the recombination rate varies over the human genome. The primary reason for this is the excitement about using *linkage disequilibrium* (LD) to map and identify human disease genes. LD mapping (also known as "association mapping") differs from traditional linkage mapping methods in that marker-disease associations are sought in populations of unrelated individuals. Such associations reflect the long evolutionary history of the chromosomal region (or the entire genome, in case of genome-wide screens), involving recombination, mutation, demographic events, and (perhaps most importantly) random reproductive success. The blocks of LD are regions, typically less than 100 kb, in which LD decreases very little with distance between markers. Between these blocks, however, LD is observed to decay rapidly with physical distance. Concomitant with the blocks of LD, these studies also find low haplotype diversity within blocks. Often, more than 90% of the chromosomes in a sample possess one of only two to five haplotypes within a block [3, 16, 15, 6, 4]. These re-

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sults also show that the SNPs within each block induce only a few distinct common haplotypes in the majority of the population, even though the theoretical number of different haplotypes for a block containing n SNPs is exponential in n .

The existence of haplotype block structure has serious implications for association-based methods for the mapping of disease genes. Conducting genome-wide association mapping studies could be much easier than it used to be based on single marker [8]. On the one hand, if a causative allele occurs within a long block of LD, then it may be difficult to localize that causative gene at a fine scale by association mapping. On the other hand, if the diversity of haplotypes within blocks is low, then common disease genes may be mapped, at least to within a haplotype block, by using far fewer markers than previously imagined [8].

Haplotype Block Designations

Our ultimate goal is to select haplotype block designations that best capture the structure within the data. Unfortunately, a consensus definition for haplotype blocks based on the LD structure has not been established thus far. However, a range of operational definitions has been used to identify haplotype-block structures, including diversity-based [2, 15, 20, 21], LD-based [6, 18], recombination-based [12, 19], and information-complexity-based [1, 13, 9] methods.

For a diversity-based test, methods can be classified into two categories: those that divide strings of SNPs into blocks on the basis of the decay of LD across block boundaries and those that delineate blocks on the basis of some haplotype-diversity measure within the blocks. Patil et al. [15] defined a haplotype block as a region in which a fraction of percent or more of all the observed haplotypes are represented at least n times or at a given threshold in the sample. The particular case implemented in our study required that in haplotype blocks, at least 80% of the observed haplotypes should be observed in at least 5% of the sample. They applied the optimization criteria outlined by Zhang et al. [20, 21] and describes a general algorithm that defines block boundaries in a way that minimizes the number of SNPs that are required to identify all the haplotype in a region.

Secondly, for a LD-based test, Gabriel et al. [6] define blocks to be a region in which a small proportion of marker pairs show evidence for historical recombination. These criteria are suggestively modified by Wall and Pritchard [18] for han-

dling haplotype data instead of unphased genotype data. Blocks are partitioned according to whether the upper and lower confidence limits on estimates of pairwise D' measure fall within certain threshold values.

The recombination-based method uses the four-gamete test of Hudson and Kaplan [12] is suggested by Wang et al. [19]; they define blocks as apparently recombination-free regions under the infinite-sites assumption.

Finally, the information complexity based method has been proposed by Anderson et al. [1]; they develop method by formalizing the task of finding block boundaries as a problem in statistical-model selection, where they apply the *minimum description length* (MDL) criterion to select the block designations that best capture the structure within the data. The MDL criterion is an application of information theory to statistical-model selection. The *description length* of a data set is a penalized negative log-likelihood, which is a function of the number and position of block boundaries. The best set of block boundaries, by the MDL criterion, is the set that achieves the shortest description length for the data. Other block-partitioning methods using the MDL criterion have been independently developed and presented at symposia. Koivisto et al. [13] report a method that uses an underlying probability model, haplotypes within blocks are clustered using k -means clustering, and the description length depends on the number of clusters within a block and how closely haplotypes within blocks cluster together. Their method is thus based on a measure of within-block haplotype diversity. Greenspan and Geiger [9] also present an MDL-based haplotype block partitioning method that has the attractive feature in that it accepts both phased haplotype data and unphased genotype data.

Ding et. al. [5] derive from a coalescent simulation under both a uniform recombination model and one that assumes recombination hotspots. There were considerable differences in haplotype information loss in the measure of entropy when the partition methods were compared under different population-genetics scenarios. Under both recombination models, the results from the LD-based definition and the recombination-based definition were more similar to each other than were the results from the diversity-based definition. This work demonstrates that when undertaking haplotype-based association mapping, the choice of haplotype-block definition and SNP selection requires careful consideration.

Terminology and Problem Definition

Abstractly, input to the haplotype blocking problem consists of m *haplotype* vectors. Each position in a vector is associated with a site of interest on the chromosome. Usually, the position in the haplotype vector has a value of 0 if it is the major allele or 1 if minor allele.

Let the *haplotype matrix* A be an $m \times n$ matrix of m observations over n markers (sites). We refer to the j th allele of observation i by A_{ij} . For simplicity, we first assume that $A_{ij} \in \{0, 1\}$. A *block*, or marker interval, $[j, k] = \langle j, j+1, \dots, k \rangle$ is defined by two marker indices $1 \leq j \leq k \leq n$. A *segmentation* is a set of non-overlapping non-empty marker intervals. A segmentation is *full* if the union of the intervals is $[1, n]$. The data matrix limited to interval $[j, k]$ is denoted by $A(j, k)$; the values of the i th observation are denoted by $A(i, j, k)$, a binary string of length $k - j + 1$. Figure 1 illustrates an example of a 7×11 0-1 haplotype matrix.

Given an interval $[j, k]$, a *diversity function*, $\delta : [j, k] \rightarrow \delta(j, k) \in \mathbb{R}$ is an evaluation function measuring the diversity of the submatrix $A(j, k)$. We say an interval $[j', k']$ is a *subinterval* of $[j, k]$, written $[j', k'] \subset [j, k]$, if $j \leq j'$ and $k' \leq k$. Note that δ -function is a *monotonic non-decreasing* function from $[1..n, 1..n]$ to the unit real interval $[0, 1]$; that is, $0 \leq \delta(j', k') \leq \delta(j, k) \leq 1$ whenever $[j', k'] \subset [j, k]$.

Given an input set of n haplotype vectors, a solution to the *Haplotype Block Selection (HBS)* Problem is a segmentation of marker intervals, revealing these non-overlapped haplotype blocks of interest in the chromosome. Here in this paper we propose several efficient algorithms related to selecting interesting haplotype blocks under different evaluation (diversity) functions that generalizes many previous results in the literatures [2, 15, 20, 21, 6, 18, 12, 19, 1, 13, 9].

Given a sample (set) of strings such that each string s_i is associated with a probability p_i ; that is $S = \{(s_1, p_1), (s_2, p_2), \dots, (s_k, p_k)\}$ such that $\sum p_i = 1$. Li [14] propose a diversity formula defined by

$$\delta_D(S) = 1 - \sum_{p_i \in S} p_i^2. \quad (1)$$

The value of $\delta_D(S)$ is the probability that two haplotype books chosen at random from S are different from each other. According to the definition, we can classify each haplotype submatrix as a sample space (multiset). For example, the submatrix $A(8, 11)$ of Figure 1 can be viewed as a sample $S =$

$\{(0001, 2/7), (0111, 2/7), (1000, 1/7), (1001, 1/7), (1100, 1/7)\}$. It follows that $\delta(S) = \delta(A(8, 11)) = \delta(8, 11) = 1 - (4 + 4 + 1 + 1 + 1)/49 = 1 - 11/49 = 38/49$. It is possible to generalize the $\delta_D(\cdot)$ function by raising the square to an arbitrary power q , then we have

$$\delta_q(S) = 1 - \sum_{p_i \in S} p_i^q. \quad (2)$$

Note that $\delta_D(S)$ is just $\delta_2(S)$. Other measurements of diversity can be obtained by choosing different diversity function; for example, to measure the information-complexity one can choose the information entropy (negative-log) function [1, 13, 9]:

$$\delta_E(S) = - \sum_{p_i \in S} p_i \log p_i. \quad (3)$$

By using appropriate diversity functions, the block selection problem can be viewed as finding a segmentation of given haplotype matrix such that the diversities of chosen blocks satisfy certain value constraint. In this paper, we consider the following problems:

Problem 1 (all-interval-diversities) *Given a haplotype matrix A , compute the all pairs block diversity values. That is, output the set $\{\delta(i, j) \mid 1 \leq i \leq j \leq n\}$.*

Using techniques of the suffix tree [10, 17], we are able to propose an $O(mn + n^2)$ time, linear proportional to the input plus output size, algorithm for computing the ALL-INTERVAL-DIVERSITIES problem.

Problem 2 (farthest-sites) *Given a haplotype matrix A and a given real diversity upper limit D , for each marker site i find its corresponding farthest right marker $j = R[i]$ so that $\delta(i, j) \leq D$. That is, output the set $\{R(i) = j \mid \delta(i, j) \leq D < \delta(i, j+1), 1 \leq i \leq j \leq n\}$. To simplify the boundary condition, we assume that $\delta(i, n+1) = \infty$.*

We show that all FARTEST-SITES can be found totally in $O(mn)$ time, linear proportional to the input size, by the the similar techniques of the suffix tree as in the ALL-PAIR-BLOCK-DIVERSITY problem.

Problem 3 (longest-block) *Given a haplotype matrix A and a diversity upper limit D , find the longest block $[i, j]$ so that $\delta(i, j) \leq D$. That is, output the $\arg \max_{i,j} \{|[i, j]| = j - i + 1 \mid [i, j] \subset [1, n] \text{ and } \delta(i, j) \leq D\}$.*

	1	2	3	4	5	6	7	8	9	10	11
h1	1	0	0	0	0	0	1	0	0	0	1
h2	1	0	1	1	0	0	0	1	1	0	0
h3	0	0	1	0	0	0	1	1	0	0	1
h4	1	1	1	0	1	0	0	0	1	1	1
h5	1	1	0	0	0	1	1	0	1	1	1
h6	0	0	0	0	0	0	1	0	0	0	1
H7	0	1	0	0	0	0	1	1	0	0	0

Figure 1: A haplotype matrix A and its corresponding submatrix $A(8, 11)$.

We show that the LONGEST-BLOCK can be found in $O(n)$ time by examining the $O(n)$ -sized farthest-sites array; that is, the the LONGEST-BLOCK can be found in linear time.

Problem 4 (k -longest-blocks) *Given a haplotype matrix A and a diversity upper limit D , find a segmentation consists of k feasible blocks such that the total length is maximized. That is, output the set $S = \{B_1, B_2, \dots, B_k\}$, with $\delta(B) \leq D$ for each $B \in S$, such that $|B_1| + |B_2| + \dots + |B_k|$ is maximized.*

We show that the k -LONGEST-BLOCKS can be found in $O(nk)$. These blocks can be identified by examining the farthest-sites array and sing the technique of sparse dynamic programming.

Problem 5 (all-intra-longest-block) *Given a haplotype matrix A , a diversity upper limit D , for all constrained interval $[i, j] \subset [1, n]$, find the corresponding longest block $B \subset [i, j]$ with $\delta(B) \leq D$. That is, output the $\arg \max_{B \subset [i, j]} \{|B| \mid B \subset [i, j] \text{ and } \delta(B) \leq D\}$.*

We show that the ALL-INTRA-LONGEST-BLOCK can be found totally $O(n^2)$ time, which is a linear time algorithm proportional to the output size.

Problem 6 (all-intra- k -longest-blocks)

Given a haplotype matrix A , a diversity upper limit D , for all constrained interval $[i, j] \subset [1, n]$, find the longest segmentation with k feasible blocks such that the total length is maximized. That is, output the set $S = \{B_1, B_2, \dots, B_k\}$, with $\delta(B) \leq D$ for each $B \in S, B \subset [i, j]$, such that $|B_1| + |B_2| + \dots + |B_k|$ is maximized.

We show that the ALL-INTRA- k -LONGEST-BLOCKS can be found totally in $O(n^2k)$ time, which is a linear time algorithm proportional to the output size.

Results

The rest of the paper is organized as the following. In Section ??, We show that all FARTHEST-SITES can be found totally in $O(mn)$ time, linear proportional to the input size. As a corollary, we show that the LONGEST-BLOCK can be found in $O(n)$ time by examining the $O(n)$ -sized farthest-sites array; that is, the the LONGEST-BLOCK can be found in linear time. In Section 2, we show that the k -LONGEST-BLOCKS can be found in $O(nk)$. These blocks can be identified by examining the farthest-sites array and sing the technique of sparse dynamic programming. We also show that the ALL-INTRA-LONGEST-BLOCK can be found totally $O(n^2)$ time, which is a linear time algorithm proportional to the output size. The problem of ALL-INTRA- k -LONGEST-BLOCKS can be found totally in $O(n^2k)$ time, which is a linear time algorithm proportional to the output size. In Section 3, we discuss approaches in dealing with missing data, and we show some hardness results concerning the minimum diversity problem.

2 Longest k Blocks with Diversity Constraints

Theorem 1 (farthest-sites) *Given an $m \times n$ haplotype matrix and a given real diversity upper limit D , for each marker site i we can find its corresponding farthest right marker $j = R[i]$ so that $\delta(i, j) \leq D < \delta(i, j + 1)$, totally in $O(mn)$ time.*

Corollary 2 (longest-block) *Given an $m \times n$ haplotype matrix and a diversity upper limit D , the longest block with diversity not exceeding D can be found in $O(mn)$ time.*

We discuss a series of problem related to finding multiple blocks with diversity constraints in the section. Given a haplotype matrix A and a

diversity upper limit D , let $S = \{B_1, B_2, \dots, B_k\}$ be a segmentation of A with $\delta(B) \leq D$ for each $B \in S$. The *length* of S is the total length of all block in S ; i.e., $\ell(S) = |B_1| + |B_2| + \dots + |B_k|$. Our objective is to find a segmentation consists of k feasible blocks such that the total length $\ell(S)$ is maximized.

Given A and D , first we consider the most general form of the problem and define the block length evaluation function

$f(k, i, j) = \max \{ \ell(S) \mid S \text{ a feasible segmentation of } A(i, j) \text{ with } k \text{ blocks} \}$

Note that the k -LONGEST-BLOCKS asks to find the value $f(k, 1, n)$. First of all, we prepare the *left farthest sites*, $L[j]$'s, for each site j of A ; according to Theorem 1, the array can be identified in $O(mn)$ time. Here we show that the answer can be found in $O(nk)$ time after the preprocessing. The idea behind the dynamic programming formula is illustrated at Figure 2.

It can be verified that

$$f(k, 1, j) = \max\{f(k, 1, j-1), f(k-1, 1, L[j] - 1) + j - L[j] + 1\}$$

That is, the k -th block of the maximal segment S in $[1, j]$ either does not include site j ; otherwise, the block $[L[j], j]$ must be the last block of S . Note that $f(k, 1, j)$ can be determined in $O(1)$ time suppose $f(k-1, 1, \cdot)$'s and $f(k, 1, 1..(j-1))$'s being ready. It follows that $f(k, 1, \cdot)$'s can be calculated from $f(k-1, 1, \cdot)$'s, totally in $O(n)$ time. Thus a computation ordering from $f(1, 1, \cdot)$'s, $f(2, 1, \cdot)$'s, $\dots$, to $f(k, 1, \cdot)$'s leads to the following result.

Theorem 3 (k -longest-blocks) *Given a haplotype matrix A and a diversity upper limit D , find a segmentation consists of k feasible blocks such that the total length is maximized can be done in $O(nk)$ time after a linear time preprocessing.*

Given A and D , the ALL-INTRA-LONGEST-BLOCK asks to find, for all constrained interval $[i, j] \subset [1, n]$, the corresponding *longest* block $B \subset [i, j]$ with $\delta(B) \leq D$. That is, the values $f(1, i, j)$'s. Similarly, we first prepare the farthest sites of A in $O(mn)$ time. Here we show that the answer can be found in $O(n^2)$ time after the preprocessing. Using similar idea as illustrated at Figure 2. It can be verified that

$$f(1, i, j) = \max\{f(1, i, j-1), (j - L[j] + 1)\}$$

That is, the longest block in $[i, j]$ either does not include site j ; otherwise, the block $[L[j], j]$ must be the longest (feasible) block within $[i, j]$. Note that $f(1, i, j)$ can be determined in $O(1)$ time

whenever $f(1, i, 1..(j-1))$'s being ready. It follows that $f(1, i, \cdot)$'s can be calculated totally in $O(n)$ time. Thus a computation ordering from $f(1, 1, \cdot)$'s, $f(1, 2, \cdot)$'s, $\dots$, to $f(1, n, \cdot)$'s leads to the following result.

Theorem 4 (all-intra-longest-block) *Given a haplotype matrix A and a diversity upper limit D , find for all constrained interval $[i, j] \subset [1, n]$, the corresponding longest block $B \subset [i, j]$ with $\delta(B) \leq D$ can be done in $O(n^2)$ time after a linear time preprocessing.*

Note that the computation time is linear proportional to the output plus input size, and thus is an optimal algorithm in regarding of time efficiency.

or all constrained interval $[i, j] \subset [1, n]$, find the longest segmentation with k *feasible* blocks such that the total length is maximized. That is, output the set $S = \{B_1, B_2, \dots, B_k\}$, with $\delta(B) \leq D$ for each $B \in S, B \subset [i, j]$, such that $|B_1| + |B_2| + \dots + |B_k|$ is maximized.

The ALL-INTRA- k -LONGEST-BLOCKS asks to find, for all constrained interval $[i, j] \subset [1, n]$, the longest segmentation with k *feasible* blocks such that the total length is maximized. Again, the problem can be solved by first prepare the farthest sites of A in $O(mn)$ time. Here we show that the answer can be found in $O(n^2k)$ time after the preprocessing. Using similar idea as illustrated at Figure 2. It can be verified that

$$f(k, i, j) = \max\{f(k, i, j-1), f(k-1, i, L[j] - 1) + j - L[j] + 1\}$$

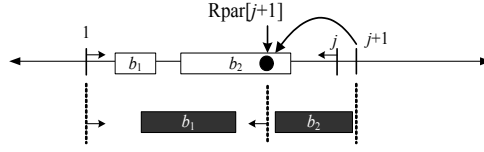
That is, the k -th block of the maximal segment S in $[i, j]$ either does not include site j ; otherwise, the block $[L[j], j]$ must be the last block of S . Note that $f(k, i, j)$ can be determined in $O(1)$ time suppose $f(k-1, i, \cdot)$'s and $f(k, i, 1..(j-1))$'s being ready. It follows that $f(k, i, \cdot)$'s can be calculated from $f(k-1, i, \cdot)$'s, totally in $O(n)$ time. Thus $f(k, \cdot, \cdot)$'s can be calculated from $f(k-1, \cdot, \cdot)$'s totally in $O(n^2)$ time. Thus a computation ordering from $f(1, \cdot, \cdot)$'s, $f(2, \cdot, \cdot)$'s, $\dots$, to $f(k, \cdot, \cdot)$'s leads to the following result.

Theorem 5 (all-intra- k -longest-blocks)

Given A, D , there exists an $O(kn^2)$ -time algorithm to find, for all constrained interval $[i, j] \subset [1, n]$, the longest segmentation with k feasible blocks such that the total length is maximized.

3 Dealing with Missing Data

We have propose several efficient algorithm in trying to identify interesting haplotype blocks in the


 Figure 2: The $O(kn)$ algorithm to compute $f(k, 1, n)$.

genome. However, in reality, some SNPs may be missing, and we may fail to distinguish two distinct haplotypes due to the ambiguity caused by missing data. These missing data may cause ambiguity in identification of an unknown haplotype sample.

Let us generalize the notion of *haplotype matrix*. Let A be an $m \times n$ matrix of m observations over n markers (sites) such that $A_{ij} \in \{0, 1, 3\}$; $A_{ij} = 3$ when the j -th SNP site of observation i is missing. One way to deal with the missing data when trying to determine the diversity of submatrix $A(j, k)$ is trying to set missing values of A_{ij} 's to either 0 or 1, such that the resulting diversity of $A(j, k)$ is minimized. It is worthy noting that the resulting diversity function is still monotonically non-decreasing. Unfortunately, as we are going to show in the following, the assignment of missing data will be very difficult since its corresponding decision problem is NP-complete.

Definition 1 (minimum-diversity) Let A be an $m \times n$ haplotype matrix with missing data such that $A_{ij} \in \{0, 1, 3\}$. The MINIMUM-DIVERSITY problem try to assign each entry $A_{ij} = 3$ to either 0 or 1 such that the resulting diversity $\delta(A)$ is minimized.

Given a haplotype matrix A with missing data, we say two rows i, j of A are *different* if there exists a column k such that $\{A_{ik}, A_{jk}\} = \{0, 1\}$; in other words, it is impossible to assign missing data of A to make the two rows identical. Two rows are *compatible* if they are not different.

Definition 2 (compatible-graph) Let A be an $m \times n$ haplotype matrix with missing data such that $A_{ij} \in \{0, 1, 3\}$. The difference graph of A , denoted by $G[A] = (V, E)$, consists of $V = \{1, 2, \dots, m\}$ as the vertex set where two vertices $ij \in E$ if and only if row i and row j are compatible.

Here we show that this combinatorial problem is intractable even when the distances measurements consist of only two different real numbers.

Theorem 6 The decision version of minimum-diversity problem is NP-complete.

Proof. The decision version of minimum-diversity problem is clearly in NP. Here we show that it is NP-hard by a reduction from the minimum clique partition (CLIPAR) [7] of simple graphs to this problem. Figure 3 illustrates the idea of reducing an instance of CLIPAR problem to the minimum diversity problem.

Given an instance of CLIPAR, $G = (V, E)$, we construct a haplotype matrix A such that the compatible graph of A is exactly G ; that is, $G[A] = G$. The construction is straightforward. Whenever $(i, j) \in E$, we construct a site, labelled as $k = (i, j)$, such that $A_{ik} = 0$ and $A_{jk} = 1$; while $A_{rk} = 3$ for each $r \neq i, j$. Furthermore, let the diversity of A is defined by

$$\delta(A) = (\text{number of different strings in } A) / m$$

It is readily verified that $\delta(A) \leq q/m$ if and only if G can be partitioned into at most q cliques. $\square$

We propose a heuristic method for dealing with missing data, the main idea is to assign the missing data $A_{i,j} = 3$ to 0 or 1 in range $[x, y]$ such that the $\delta(A(x, y))$ is minimized. By reducing the patterns of haplotype as few as possible, the assigned haplotype patterns would lead to small diversity score.

The heuristic algorithm consists of three phases. First, we partition $A(x, y)$ into two sets. We call the set of strings with missing data as T , and those strings without missing data are called S . Strings in S can be grouped into k patterns (distinct strings) $p_1, \dots, p_k$. Second, for strings in T , if there is a string $t \in T$ is compatible with some strings $p_i \in S$, then assign string t to p_i such that the string number of p_i is the largest in the set $\{p_1, \dots, p_k\}$. Note that the patterns of haplotype will not increase and the assignment leads to the lowest diversity score possible if such compatible matching is possible.

The third phase deals with the remain strings, called R , independently. The idea is to assigning those missed SNP sites of R such that the resulted

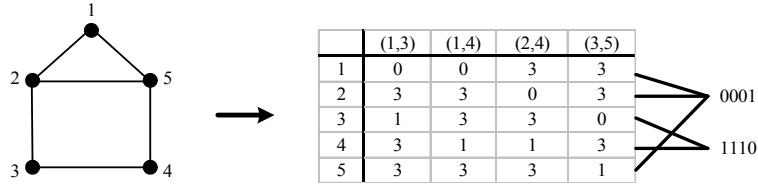


Figure 3: A given graph G and the constructed minimum diversity instance.

haplotype patterns are as few as possible (with low diversity.) There are two steps in the phase. First, order the strings in set R , $\langle r_1, \dots, r_n \rangle$, by the increasing order of number of missing sites; the string containing the least missing data is placed at the beginning. Second, compare each string r_i , for $2 \leq i \leq |R|$, to other above strings r_j , $j < i$. We *consolidate* r_i and r_j if the two strings are compatible and delete r_i from the ordered list R . Repeat the step until all strings are not compatible with each other. Figure 4 illustrates the algorithm of the third phase of our method.

Note that missing data may still exist in the haplotype matrix after the missing data process; e.g., some strings are never compatible with the others or the values of one specific column are all missing data. It's worthy noting that the situations do not influence the computation of diversity because two strings are always different no matter how we assign the missing data and the values of one specific column with all missing data are the same.

4 Preliminary Experimental Results

We use the haplotype data used in previous literatures [3, 15, 11, 4] and download several sets of haplotype data from NCBI and other public databases to evaluate the performance of our block selection algorithms. Chromosome 21 spans 44.6Mbps long, the SNP data we adopt locates at the region from the beginning of chromosome 21 to 28.5Mbps, and the haplotype we use to evaluate partition results is in the range from 14.3Mbps to the end of this chromosome. We focus on the overlapped region and evaluate the performance for different block selection results. The overlapped region 9911 SNPs and 840 haplotypes. We also try to deal with missing data by heuristic related to minimum diversity strategy as discussed in Section 3.

Based on these obtained data, we have de-

veloped a web-based system, mostly consists of a list of PHP and perl CGI-scripts together with several C programs for selecting haplotype blocks and analyzing the haplotype diversity. The system is still under development and the web system is freely accessible at http://bioinfo.cs.pu.edu.tw/PU_HapSel.htm. Some preliminary results, including the selection of different diversity functions as well as choosing meaningful diversity constraints, in using our tools can also be found in web system. Our web system also provide the haplotype generator. The users can input the size of haplotype matrix and the ratio of value 0, our system will generate a haplotype matrix for user to analysis automatically.

5 Concluding Remarks

By using appropriate diversity functions, the block selection problem can be viewed as finding a segmentation of given haplotype matrix such that the diversities of chosen blocks satisfy certain value constraint. In this paper, we propose several efficient combinatorial algorithms related to selecting interesting haplotype blocks under different diversity functions.

Using several dynamic programming deduction, we show in Theorem 1 that the farthest sites problems problem can be computed in $O(mn)$ time. As a corollary, we also show that the longest block can also be found in $O(mn)$ time. In Theorem 3, we show that finding longest k -block segmentation with diversity constraints can be done in $O(nk)$ time. In Theorem 4, we show that finding the longest blocks with each constrained interval can be done in $O(n^2)$ time. In Theorem 5, we show that finding all longest k -block segmentation with diversity constraints can be done in $O(n^2k)$ time. In Theorem 6, we prove that the minimum diversity problem is NP-complete.

We need to point out that these time-efficiency results of our algorithms can be applied in many different definitions of diversity functions; there we

LowDiv(R)

Input: The incompatible remaining strings in $A(x, y)$ with missing data.

Output: Resolving missing SNP sites in R so that the remaining patterns contain low diversity.

```

1 Let  $R = \{r_1, r_2, \dots, r_{|R|}\}$  be rows containing missing data.
   $\triangleright$   $r_i$ 's are listed in increasing order of number of missed SNP sites.
2 for  $i \leftarrow 2$  to  $|R|$  do
3   for  $j \leftarrow 1$  to  $i - 1$  do
4     if COMPATIBLE( $r_i, r_j$ ) then
5       CONSOLIDATE( $r_i, r_j$ )
6       delete  $r_j$  from  $R$ 
```

Figure 4: The third phase of our heuristic method for dealing with missing data.

assume that the computation of the diversity of submatrix $A(j, k')$ can be obtained from a shorter prefix of $A(j, k)$, with $k < k'$, in time proportional to the size of changing events. It is not hard to verify that the diversity functions listed in Eq. (1), (2), (3), as well as many other different diversity functions, possess this property.

References

- [1] Eric C. Anderson and John Novembre. Finding haplotype block boundaries by using the minimum-description-length principle. *Am. J. of Human Genetics*, 73:336–354, 2003.
- [2] D. Clayton. Choosing a set of haplotype tagging SNPs from a larger set of diallelic loci. *Nature Genetics*, 29(2), 2001.
- [3] M. Daly, J. Rioux, S. Schafner, T. Hudson, and E. Lander. Highresolution haplotype structure in the human genome. *Nature Genetics*, 29:229–232, 2001.
- [4] E. Dawson, G. Abecasis, et al. A first-generation linkage disequilibrium map of human chromosome 22. *Nature*, 418:544–548, 2002.
- [5] Keyue Ding, Kaixin Zhou, Jing Zhang, Joanne Knight, Xuegong Zhang, and Yan Shen. The effect of haplotype-block definitions on inference of haplotype-block structure and htSNPs selection. *Molecular Biology and Evolution*, 22(1):148–159, 2005.
- [6] S. B. Gabriel, S. F. Schaffner, H. Nguyen, et al. The structure of haplotype blocks in the human genome. *Science*, 296(5576):2225–2229, 2002.
- [7] M.R. Garey and D.S. Johnson. *Computers and Intractability – A Guide to the Theory of NP-Completeness*. Freeman, New York, 1979.
- [8] D.B. Goldstein. Islands of linkage disequilibrium. *Nature Genetics*, 29:109–111, 2001.
- [9] G. Greenspan and D. Geiger. Model-based inference of haplotype block variation. In *Seventh Annual International Conference on Computational Molecular Biology (RECOMB)*, 2003.
- [10] D. Gusfield. *Algorithms on Strings, Trees and Sequences: Computer Science and Computational Biology*. Cambridge University Press, 1997.
- [11] L. Helmuth. Genome research: Map of the human genome 3.0. *Science*, 293(5530):583–585, 2001.
- [12] R. R. Hudson and N. L. Kaplan. Statistical properties of the number of recombination events in the history of a sample of dna sequences. *Genetics*, 111:147–164, 1985.
- [13] M. Koivisto, M. Perola, R. Varilo, W. Henrich, J. Ekelund, M. Lukk, L. Peltonen, E. Ukkonen, and H. Mannila. An mdl method for finding haplotype blocks and for estimating the strength of haplotype block boundaries. In *8th Pacific Symposium on Biocomputing (PSB)*, pages 502–513, 2003.
- [14] W.H. Li and D. Graur. *Fundamentals of Molecular Evolution*. Sinauer Associates, Inc, 1991.

- [15] N. Patil, A. J. Berno, D. A. Hinds, et al. Blocks of limited haplotype diversity revealed by high resolution scanning of human chromosome 21. *Science*, 294:1719–1723, 2001.
- [16] D. Reich, M. Cargill, E. Lander, et al. Linkage disequilibrium in the human genome. *Nature*, 411:199–204, 2001.
- [17] Esko Ukkonen. On-line construction of suffix trees. *Algorithmica*, 14(3):249–260, 1995.
- [18] J.D. Wall and J.K Pritchard. Haplotype blocks and linkage disequilibrium in the human genome. *Nature Reviews Genetics*, 4(8):587–597, 2003.
- [19] N. Wang, J.M. Akey, K. Zhang, R. Chakraborty, and L. Jin. Distribution of recombination crossovers and the origin of haplotype blocks: the interplay of population history, recombination, and mutation. *Am. J. Human Genetics*, 71:1227–1234, 2002.
- [20] K. Zhang, M. Deng, T. Chen, M.S. Waterman, and F. Sun. A dynamic programming algorithm for haplotype block partitioning. In *The National Academy of Sciences*, volume 99, pages 7335–7339, 2002.
- [21] K. Zhang, Z.S. Qin, J.S. Liu, T. Chen T, M.S. Waterman, and F. Sun. Haplotype block partitioning and tag snp selection using genotype data and their applications to association studies. *Genome Res.*, 14(5):908–916, 2004.